

AI531C — Week 4 Pre-Class Activity (PCM-4: The Noise Line-Up)

Prepares: Lecture 10 (Degradation model & noise processes, Tue 25 Aug) · Tutorial T4 (Tue 25 Aug).
Wed 26 Aug is a holiday.

1 · Topic

Unit II opens: the degradation model $g = h * f + \eta$, and learning to recognise noise by its fingerprint before choosing any cure.

2 · Why you are preparing this

Lecture 10 is diagnosis before treatment. If you arrive able to TELL noises apart by histogram shape, class time goes to the harder question — which physics produced each fingerprint, and what that means for the filters of Lecture 11.

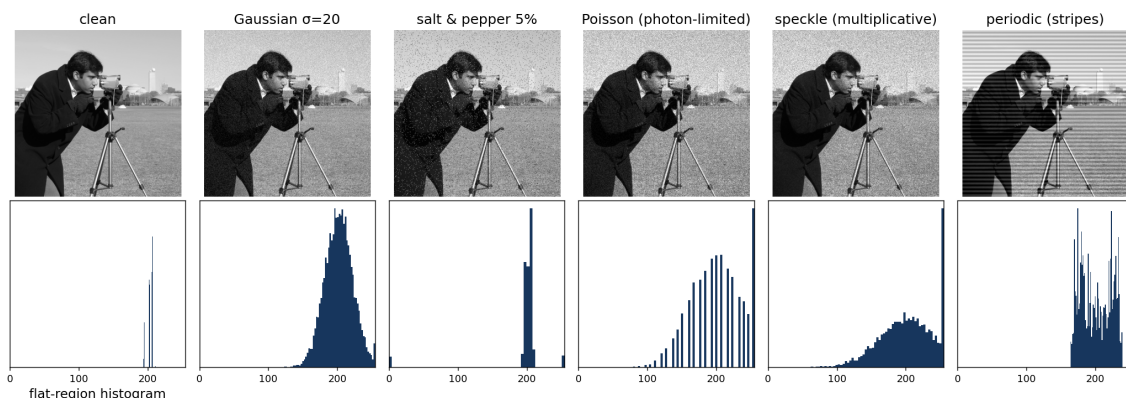
3 · Read (12-15 min)

From the Comprehensive Self-Contained Lecture Notes:

- Lecture 10 chapter, the sections on the degradation model and on the five noise processes (about 3 pages);
- revisit ONE page: Lecture 9's summary (the frequency view returns as h 's fingerprint).

Optional depth: GW4 Ch. 5, Secs. 5.1-5.2 (the noise-PDF figures).

4 · Interpret one figure (4 min)



Source: course figure, Lecture 10 (noise line-up with flat-patch histograms).

In two sentences: which panel is salt-and-pepper and which is Gaussian, and what feature of the FLAT-PATCH histogram gives each away?

5 · One small calculation (5 min)

An image is corrupted with zero-mean Gaussian noise, $\sigma = 20$ (8-bit scale). Compute the MSE this noise contributes on average, and the PSNR it implies (255 peak). One decimal place.

6 · Predict (4 min)

Predict: for Poisson (photon) noise, will the dark regions or the bright regions of a photo look noisier — and is that the same answer as for Gaussian noise?

7 · Inspect this fragment (3 min)

```
noisy = img + rng.normal(0, 20, img.shape)
patch = noisy[flat_y, flat_x] # a FLAT wall region
plt.hist(patch.ravel(), bins=50) # why a flat patch?
```

Question: why must the histogram be taken on a FLAT patch for the fingerprint to show? What would a textured patch mix in?

8 · Bring one question to class

One question about noise or the degradation model — or one photo situation where you cannot tell whether the problem is noise, blur, or both.

Submit: items 4-5 and your prediction — half a page maximum, before Tue 25 Aug, 9:00 AM.

Expected time: 30–35 minutes. Counts toward the weekly preparation portfolio (best 10 of PCM-2...PCM-14, completion-based micro-credit per the lesson plan). The short per-lecture primers in the course folder are optional warm-ups; only THIS card is submitted.